ASSIGNMENT 4

Assignment Title - Create any interactive data visualization dashboard for any real-life application like weather dashboard. Visualize data using charts and graphs.

1. **Objectives -**
   a. To design an interactive **Weather Monitoring Dashboard** using data visualization techniques.
   b. To implement different types of charts such as **Line Chart and Bar Charts**.
   c. To use the **D3.js library** for creating dynamic and animated graphs.
   d. To visualize real-life weather parameters like temperature, humidity, and rainfall.
   e. To understand how data can be represented graphically for better analysis.

2. **Outcomes -**
   After completing this assignment, I was able to:

- Create a fully interactive dashboard using HTML, CSS, and D3.js.
- Display weather data using **line graphs and bar charts**.
- Implement city-based filtering using a dropdown menu.
- Calculate average temperature and total rainfall dynamically.
- Understand the importance of data visualization in real-world applications.

3. **Theory Concept in brief -**

1. Data Visualization

Data visualization is the graphical representation of data using charts and graphs.
It helps in understanding patterns, trends, and comparisons easily.

Examples:

- Line chart → Shows trends over time
- Bar chart → Shows comparisons
- Pie chart → Shows proportions

2. D3.js (Data-Driven Documents)

D3.js is a powerful JavaScript library used to create interactive and animated data visualizations in web browsers.

It allows:

- Binding data to HTML/SVG elements
- Creating scales and axes
- Animating graphs
- Updating charts dynamically

3. Weather Dashboard Concept

A weather dashboard typically displays:

- Daily temperature trends
- Humidity levels
- Rainfall data
- Comparison between multiple cities

In this assignment:

- Line Chart → Displays weekly weather trends (Temperature, Humidity, Rainfall).
- Bar Chart 1 → Displays average temperature of all cities.
- Bar Chart 2 → Displays total weekly rainfall.
- Dropdown menu allows selecting a specific city.

4. Dataset used -

A manually created dataset was used inside the JavaScript file.

The dataset contains weather information for the following cities:

- Pune
- Mumbai
- Delhi
- Nanded
- Nagpur
- Hyderabad
- Bengaluru
- Chennai
- Jaipur

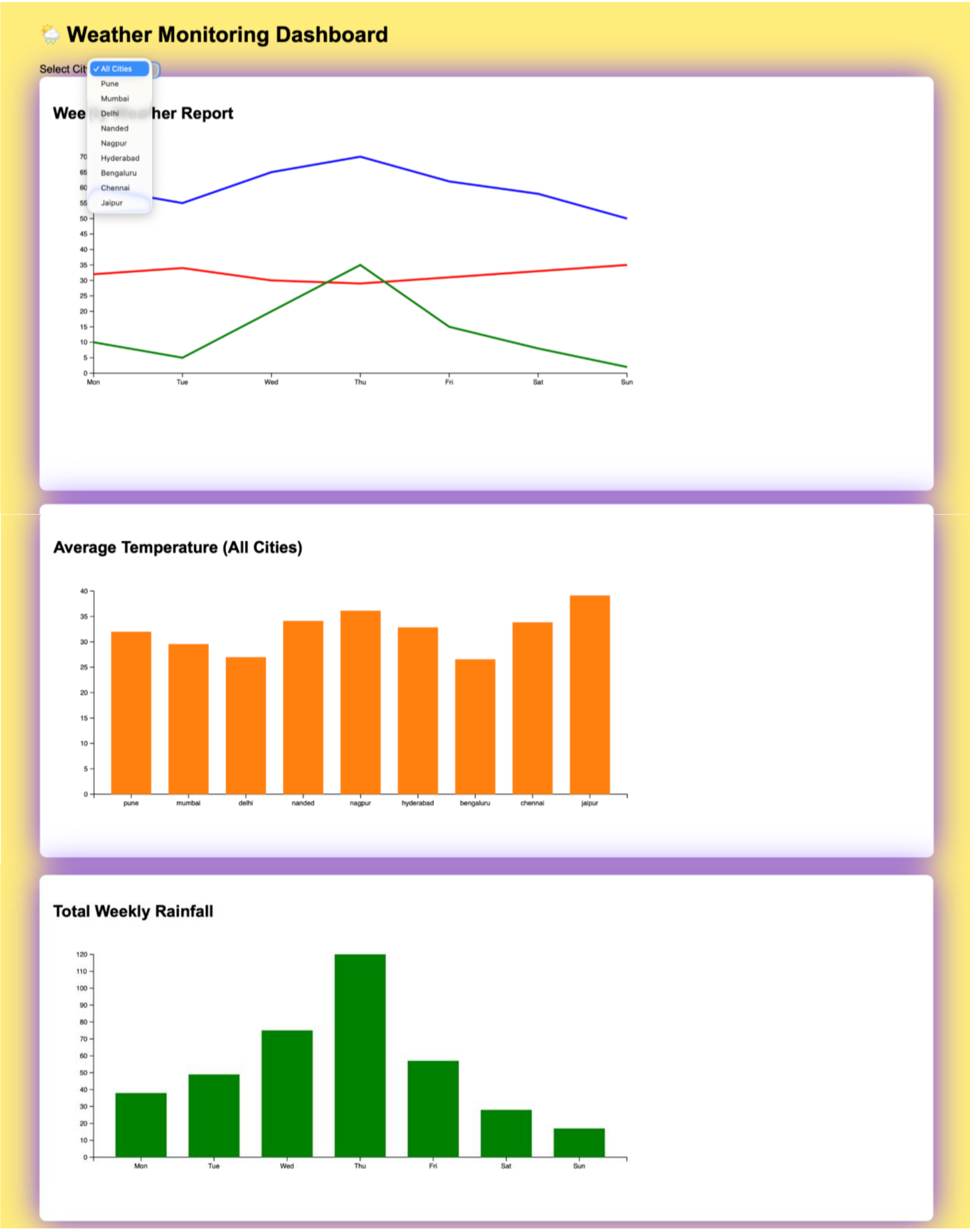
Each city includes:

- Day (Mon–Sun)
- Temperature
- Humidity
- Rainfall

No external dataset was used.

5. Program code with output screenshot -

Assignment Code : [Link]



The dashboard includes:

1. Interactive dropdown for city selection
2. Animated line chart for weekly weather report
3. Bar chart for average temperature
4. Bar chart for total weekly rainfall
5. Clean UI with styled cards

6. Conclusion -

In this assignment, I successfully developed an interactive Weather Monitoring Dashboard using HTML, CSS, and D3.js.
The dashboard visually represents weather data through dynamic charts, making it easier to analyze trends and comparisons between cities.

By implementing line and bar charts with animation and filtering options, I understood how real-world dashboards are designed.
This assignment improved my understanding of data visualization, D3.js library usage, and dynamic web-based graphical representation.